EXTENSIONS OF MODULES OF THE STANDARD PRESENTATION FORM ARE AGAIN OF THAT FORM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites a situation and an equation from the same chapter, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Refinement of a presentation towards constructible sheaves. Stacks Project, Modules on Sites, Lemma lemma-change-presentation-towards-constructibles (tag 09B6). Supplies the refinement step that lets the two presentations be combined in the basis.
- The quasi-compact objects situation. Stacks Project, Modules on Sites, Situation situation-quasi-compact-objects (tag 09AZ). Provides the basis and the enough-objects hypotheses under which the refinement is available.

Source reference. Original reference 0937: sources/stacks-chow/tags.txt, source lines 11505–11505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0938: sources/stacks-chow/tags.txt, source lines 11506–11506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0939: sources/stacks-chow/tags.txt, source lines 11507–11507. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 093A: sources/stacks-chow/tags.txt, source lines 11508–11508. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 093D: sources/stacks-chow/tags.txt, source lines 11511–11511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 093F: sources/stacks-chow/tags.txt, source lines 11513–11513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0G1W: sources/stacks-chow/tags.txt, source lines 20036–20036. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 0937 assume (0938), (0939), and (093A) hold. Let $\mathcal{O}$ be a sheaf of rings. Then an extension of modules as in (093D) is another module as in (093D).*

Proof. Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ be a short exact sequence of $\mathcal{O}$-modules with $\mathcal{F}_1$ and $\mathcal{F}_3$ as in (093D). Choose presentations

$$\bigoplus A_{V_j} \to \bigoplus A_{U_i} \to \mathcal{F}_1 \to 0 \quad \text{and} \quad \bigoplus A_{T_j} \to \bigoplus A_{W_i} \to \mathcal{F}_3 \to 0$$

In this proof the direct sums are always finite, and we write $A_U = j_{U!}\mathcal{O}_U$ for $U \in \mathcal{B}$. Since $\mathcal{F}_2 \to \mathcal{F}_3$ is surjective, we can choose coverings $\{W_{ik} \to W_i\}$ with $W_{ik} \in \mathcal{B}$ such that $A_{W_{ik}} \to \mathcal{F}_3$ lifts to a map $A_{W_{ik}} \to \mathcal{F}_2$. By Lemma 093F we may replace our collection $\{W_i\}$ by a finite subcollection of the collection $\{W_{ik}\}$ and assume the map $\bigoplus A_{W_i} \to \mathcal{F}_3$ lifts to a map into $\mathcal{F}_2$. Consider the kernel

$$\mathcal{K}_2 = \mathrm{Ker}(\bigoplus A_{U_i} \oplus \bigoplus A_{W_i} \longrightarrow \mathcal{F}_2)$$

By the snake lemma this kernel surjects onto $\mathcal{K}_3 = \mathrm{Ker}(\bigoplus A_{W_i} \to \mathcal{F}_3)$. Thus, arguing as above, after replacing each T_j by a finite family of elements of $\mathcal{B}$ (permissible by Lemma 0G1W) we may assume there is a map $\bigoplus A_{T_j} \to \mathcal{K}_2$ lifting the given map $\bigoplus A_{T_j} \to \mathcal{K}_3$. Then $\bigoplus A_{V_j} \oplus \bigoplus A_{T_j} \to \mathcal{K}_2$ is surjective which finishes the proof. □